

## Course Syllabus

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| <b>STI VISION:</b><br>To be the leader in innovative and relevant education that nurtures individuals to become competent and responsible members of society                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>STI INSTITUTIONAL OUTCOMES:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
| <b>STI MISSION:</b><br>We are an institution committed to provide knowledge through the development and delivery of superior learning systems.<br><br>We strive to provide optimum value to all our stakeholders - our students, our faculty members, our employees, our partners, our shareholders, and our community.<br><br>We will pursue this mission with utmost integrity, dedication, transparency, and creativity. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Character (IO1):</b> An STler is a person of character. An STler takes responsibility for his/her actions, treats people with respect, and lives with integrity.<br><br><b>Critical thinker (IO2):</b> An STler is a critical thinker. An STler challenges and analyses all information through sound questioning and is unafraid to push for creative ideas.<br><br><b>Communicator (IO3):</b> An STler communicates to understand and be understood. An STler discerns the value of information read or heard and effectively expresses his/her own emotions when sharing information, may it be spoken or written.<br><br><b>Change-adept (IO4):</b> An STler is change-adept. An STler can adjust, adapt, and reinvent continuously to changing circumstances. An STler believes in letting go of the old and embracing the new to achieve his/her fullest potential. |  |
| <b>SERIAL NUMBER: IT2012</b>                                                                                                                                                                                                                                                                                                                                                                                                | <b>SUBJECT TITLE: GAME DEVELOPMENT/GAME PROGRAMMING</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>CREDIT:</b> 2 lec, 1 lab (2 hrs. lec and 3 hrs. lab per week)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| <b>COURSE DESCRIPTION:</b>                                                                                                                                                                                                                                                                                                                                                                                                  | This course gives an overview of the game development process from conception to production. Game development includes gameplay, storytelling, challenges, and basic interface design, which includes interface design, information design, and world interaction.                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <b>PREREQUISITE:</b>                                                                                                                                                                                                                                                                                                                                                                                                        | Programming Languages (BSIT)/Computer Programming 3 (BSCS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <b>COURSE OUTCOMES:</b>                                                                                                                                                                                                                                                                                                                                                                                                     | After successful completion of this course, the student should be able to: (CSPO01-PI1, CSPO02-PI2, CSPO03-PI3, CSPO06-PI4)<br>CO1. Evaluate the game development process from concept to production;<br>CO2. Examine the concepts behind game development; and<br>CO3. Apply the game concepts and techniques in creating a game.                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <b>MANDATED BOOK:</b>                                                                                                                                                                                                                                                                                                                                                                                                       | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <b>REFERENCES:</b>                                                                                                                                                                                                                                                                                                                                                                                                          | 1. Buttfield-Addison, P., Manning, J., & Nugent, T. (2019) <i>Unity game development cookbook</i> . Sebastopol: O'Reilly Media, Inc.<br>2. Caldwell, C. (2018) <i>Story structure and development</i> . CRC Press.<br>3. Casanova, M., Doran, J. (2018) <i>Game development patterns and best practices: better games, less hassle</i> . Packt Publishing Ltd.<br>4. Lanzinger, F. (2020) <i>2D Game development with unity</i> . CRC Press.<br>5. Novak, J. (2018). <i>Game development essentials: An introduction (3<sup>rd</sup> ed.)</i> . New York: Delmar, Cengage Learning. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <b>COURSE REQUIREMENTS:</b>                                                                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"><li>• Class participation (Recitation, Seatwork, Quizzes, etc.)</li><li>• Task performance (eLMS Activities, Laboratory Exercises, Projects, etc.)</li><li>• Major examination</li></ul>                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <b>GRADING SYSTEM:</b>                                                                                                                                                                                                                                                                                                                                                                                                      | The following percentage distribution shall be followed:<br>Prelims 20%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |

|               |            |
|---------------|------------|
| Midterms      | 20%        |
| Pre-finals    | 20%        |
| <u>Finals</u> | <u>40%</u> |
|               | 100%       |

The following are the periodical grade components for this course:

|                          |            |
|--------------------------|------------|
| Class Participation      | 20%        |
| Task Performance         | 30%        |
| <u>Major Examination</u> | <u>50%</u> |
|                          | 100%       |

*We'd be glad to hear from you. For questions or feedback on this course, feel free to email us through [student.feedback@sti.edu](mailto:student.feedback@sti.edu)*

## Course Outline

| Learning Objectives (LO) |                                                                           | Week | LEC Hours | LAB Hours | TOPICS                                                                               | Teaching and Learning Activities        | Assessment Tasks          |
|--------------------------|---------------------------------------------------------------------------|------|-----------|-----------|--------------------------------------------------------------------------------------|-----------------------------------------|---------------------------|
| 1                        | Differentiate game platforms and distinguish their characteristics. (CO1) | 1-2  | 4         | 3         | <b>Platforms and Player modes</b><br><br>Platforms<br>Time Intervals<br>Player Modes | Lecture                                 | 01 eLMS Activity 1        |
| 2                        | Describe each classification of time interval and player modes. (CO2)     |      |           |           |                                                                                      | Demonstration                           | 01 Laboratory Exercise 1  |
| 3                        | Create an asset for microgame. (CO3)                                      |      |           |           |                                                                                      |                                         |                           |
| 4                        | Identify the goals and purposes of games. (CO1)                           | 3-4  | 4         | 6         | <b>Goals and Genres</b><br><br>Goals<br>Genres                                       | Lecture                                 | 02 eLMS Quiz 1            |
| 5                        | Compare the classifications of game genres. (CO2)                         |      |           |           |                                                                                      | Demonstration                           | 02 Laboratory Exercise 1  |
| 6                        | Create and modify an asset for an FPS microgame. (CO3)                    |      |           |           |                                                                                      |                                         | 02 Task Performance 1     |
|                          |                                                                           | 5    |           |           | <b>PRELIMINARY EXAMINATION</b>                                                       |                                         | <b>Pen-and-Paper Test</b> |
| 7                        | Determine the reasons for playing games. (CO1)                            | 6-7  | 4         | 6         | <b>Player Elements</b><br><br>Player Motivation<br>Game Market                       | Lecture                                 | 03 eLMS Activity 1        |
| 8                        | Discuss the advantages of understanding the game market. (CO2)            |      |           |           |                                                                                      | Demonstration<br><br>03 eLMS Activity 1 | 03 Laboratory Exercise 1  |

| Learning Objectives (LO) |                                                                                          | Week  | LEC Hours | LAB Hours | TOPICS                                                                                                                                         | Teaching and Learning Activities | Assessment Tasks                                                                |
|--------------------------|------------------------------------------------------------------------------------------|-------|-----------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|---------------------------------------------------------------------------------|
| 9                        | Describe the three-act plot structure and story elements. (CO2)                          | 8-9   | 4         | 6         | <b>Story and Character Development</b><br><br>Traditional Story Structure<br>Story Elements<br>Classic Character Archetypes<br>Game Characters | Lecture<br><br>Demonstration     | 04 eLMS Activity 1<br><br>04 Laboratory Exercise 1<br><br>04 Task Performance 1 |
| 10                       | Differentiate the character archetypes and game characters. (CO2)                        |       |           |           |                                                                                                                                                |                                  |                                                                                 |
| 11                       | Create a game pre-production. (CO3)                                                      |       |           |           |                                                                                                                                                |                                  |                                                                                 |
|                          |                                                                                          | 10    |           |           | <b>MIDTERM EXAMINATION</b>                                                                                                                     |                                  | Pen-and-Paper Test                                                              |
| 12                       | Identify the impact of gameplay in game development. (CO1)                               | 11-12 | 4         | 6         | <b>Gameplay</b><br><br>Fundamentals<br>Challenges & Game Goals<br>Balance                                                                      | Lecture<br><br>Demonstration     | 05 eLMS Activity 1<br><br>05 Laboratory Exercise 1                              |
| 13                       | Distinguish the different game challenges and their specific goals. (CO2)                |       |           |           |                                                                                                                                                |                                  |                                                                                 |
| 14                       | Create a gameplay mode for microgames. (CO3)                                             |       |           |           |                                                                                                                                                |                                  |                                                                                 |
| 15                       | Determine the impact of level design and structure in game development. (CO2)            | 13    | 2         | 3         | <b>Levels</b><br><br>Level Design<br>Structure<br>Time<br>Space                                                                                | Lecture<br><br>Demonstration     | 06 eLMS Activity 1<br><br>06 Quiz 1<br><br>06 Task Performance 1                |
| 16                       | Distinguish the effects of time and space in the physical environment of the game. (CO2) |       |           |           |                                                                                                                                                |                                  |                                                                                 |
| 17                       | Design a gameplay level. (CO3)                                                           |       |           |           |                                                                                                                                                |                                  |                                                                                 |
|                          |                                                                                          | 14    |           |           | <b>PRE-FINAL EXAMINATION</b>                                                                                                                   |                                  | Pen-and-Paper Test                                                              |

| Learning Objectives (LO) |                                                | Week  | LEC Hours | LAB Hours | TOPICS                               | Teaching and Learning Activities | Assessment Tasks     |
|--------------------------|------------------------------------------------|-------|-----------|-----------|--------------------------------------|----------------------------------|----------------------|
| 18                       | Create a working prototype game. (CO3)         | 15-17 | 6         | 6         | Project Development and Presentation | Group Activity                   | Project Development  |
| 19                       | Present the final output of the project. (CO3) |       |           |           |                                      | Consultation                     | Project Presentation |
|                          |                                                | 18    |           |           | FINAL EXAMINATION                    |                                  | Pen-and-Paper Test   |

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| <b>PREPARED BY:</b><br>Carlo E. Magpili<br>Kierl Christian Roi D. Panares | <b>VERIFIED BY:</b><br>Alyanna R. Tobias, Ed.M., LPT | <b>REVIEWED BY:</b><br>Beronika A. Peña | <b>APPROVED BY:</b><br>Fernando T. Dantes III, MIT |  |
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